

CFD-HW5

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Outline

In this assignment, I was tasked with modeling the air flow around an airfoil at various angles of attack in a wind tunnel. Two dimensional analysis will be carried out on the model.

Setup

The setup involves an airfoil with a length of 1 m inside an rectangular air tunnel with a length of 80 m and a height of 8 m. The (vertically centered) airfoil is positioned 20 m from the inlet. We will allow the angle of attack to be $\alpha = \{0^\circ, 10^\circ, 20^\circ\}$ and the initial air speed to be $v = \{102.9, 171.5\}$ m/s (noting Mach 1 = 343 m/s) at STP conditions. Overall, there are six configurations to analyze.

Before getting my hands dirty, upon opening the provided CAD model for the airfoil, I noticed it was scaled down such that each meter corresponded to an inch in the model. So I opened up the model in Autodesk Inventor and scaled it up (uniformly) by a factor of 39.3701. Now the airfoil had a length of 1 m and an (approximate) maximum height of 0.12 m. By default, the width of the airfoil is also 1 m.

Preliminary Calculations

The lift coefficient can be approximated as:

$$C_L = 2\pi\alpha \quad (\text{assuming } \alpha \text{ is measured in radians})$$

The drag coefficient can be approximated as:

$$C_D = 2\sin^2\alpha$$

The characteristic area of the airfoil is given by:

$$A = (\text{length})(\text{width}) = (1\text{ m})(1\text{ m}) = 1\text{ m}^2$$

The dynamic pressure (difference) across the airfoil is given by:

$$q = \frac{1}{2}\rho v^2 \quad (\text{where } \rho = 1.225\text{ kg/m}^3)$$

The lift and drag forces are thus given as such:

$$F_L = C_L q A \quad \text{and} \quad F_D = C_D q A$$

Table 1: Mach 0.3 Calc. Tabulation

α	F_L	F_D
0°	0 N	0 N
10°	7150 N	390 N
20°	14200 N	1500 N

Table 2: Mach 0.5 Calc. Tabulation

α	F_L	F_D
0°	0 N	0 N
10°	19800 N	1080 N
20°	39400 N	4140 N

Obviously the lift & drag calculation are dead wrong for 0° , but that's because the airfoil was assumed to be a thin plate. However, because the (max) height to length ratio is (about) $0.12/1 = 0.12 < 1$, the calculation shouldn't be too off. This is almost definitely going to blow up in my face...

Geometric CAD Work

With all of that out of the way it was time to implement the airfoil geometry within the wind tunnel. Unfortunately, the proprietary CAD tools inside Ansys have a UI akin to horse scat, and so, with the help of my friend David Brokhin, the wind tunnel geometry was successfully cut out of the wind tunnel volume at the prescribed angles of attack inside of Autodesk Inventor. I was then able to import the geometries into Design Modeler as a gut check:

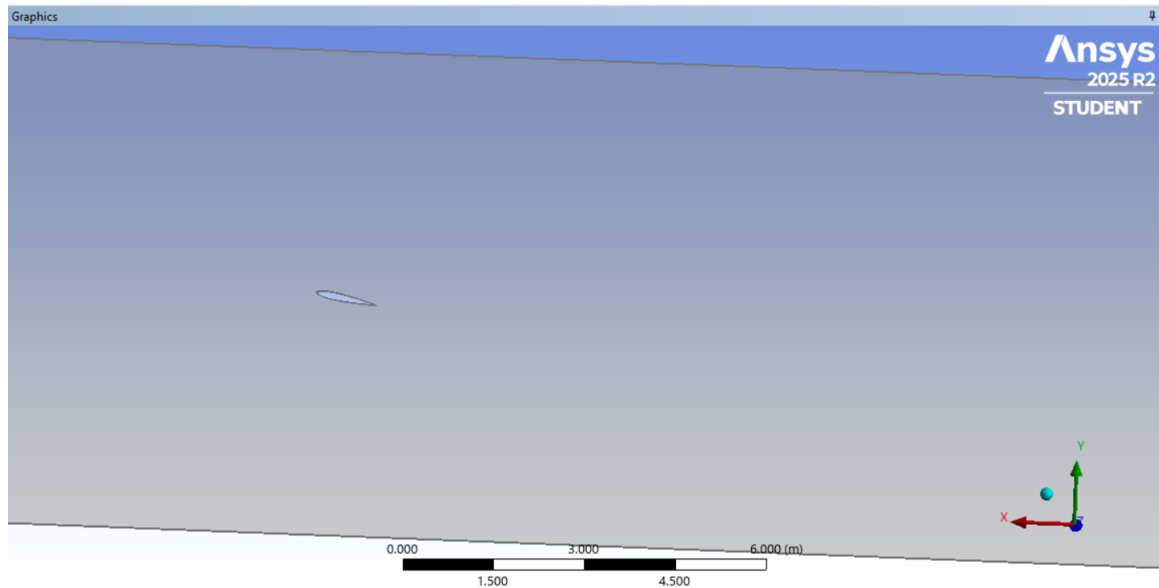


Figure 0: Preliminary Geometry Work ($\alpha = 10^\circ$)

Case 1: $\alpha = 0^\circ$

To start, we need a mesh. Provided a baseline cell size of 0.2 m, in conjunction with inflation (around the airfoil) with 2 layers and edge sizing (0.001 m) to capture the important detail, the following mesh was achieved, and it's actually not too bad:

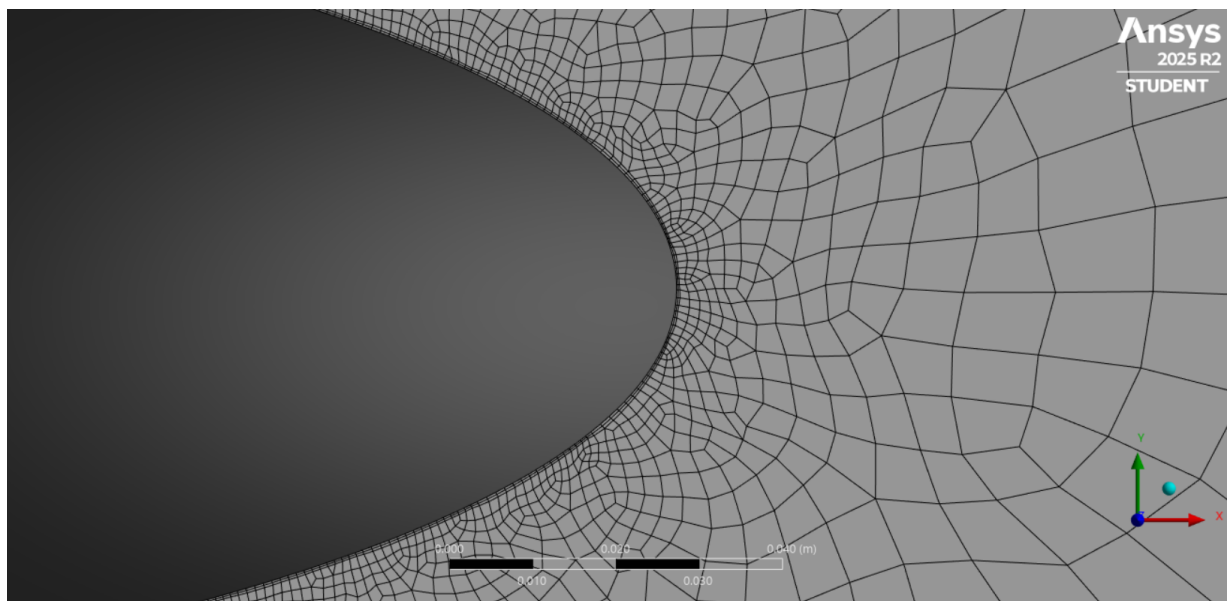


Figure 1.0.0: Mesh Work ($\alpha = 0^\circ$)

The mesh was plugged into the CFD solver (coupled, $k - \omega$, 300 iterations) with the appropriate boundary conditions for $v = 102.9$ m/s, and the following results (below) were acquired. Before looking, it will be noted here that all the force reports moving forward will be tabulated at the very end for conciseness with the exception of Case 1:

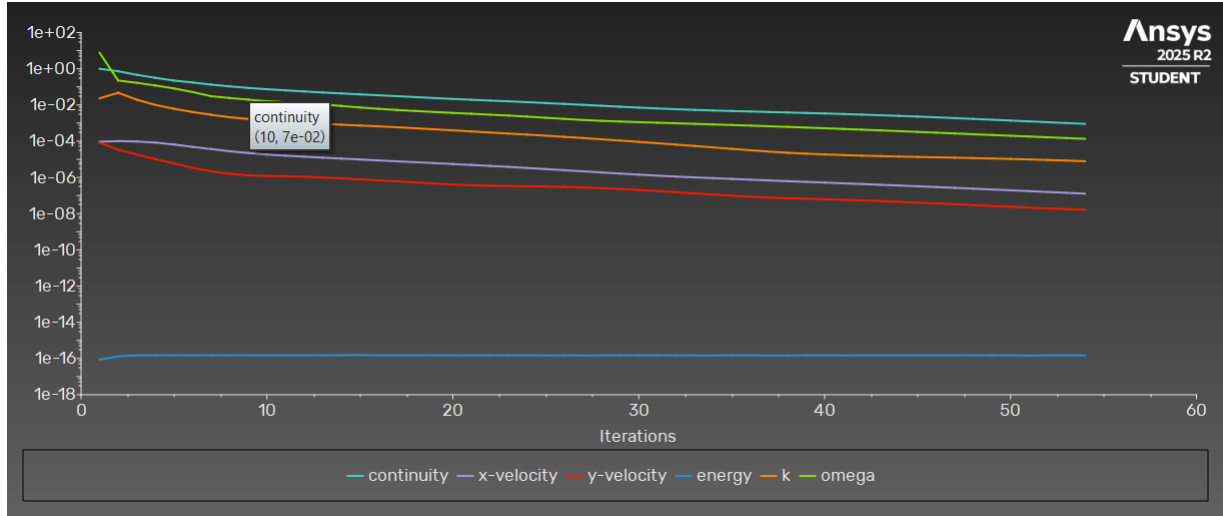


Figure 1.1.1: Residuals ($\alpha = 0^\circ$ and $v = 102.9$ m/s)

I don't know about you, but from my position, that energy residual is way below 10^{-6} , so I'd say Navier-Stokes converged quite well over the mesh.

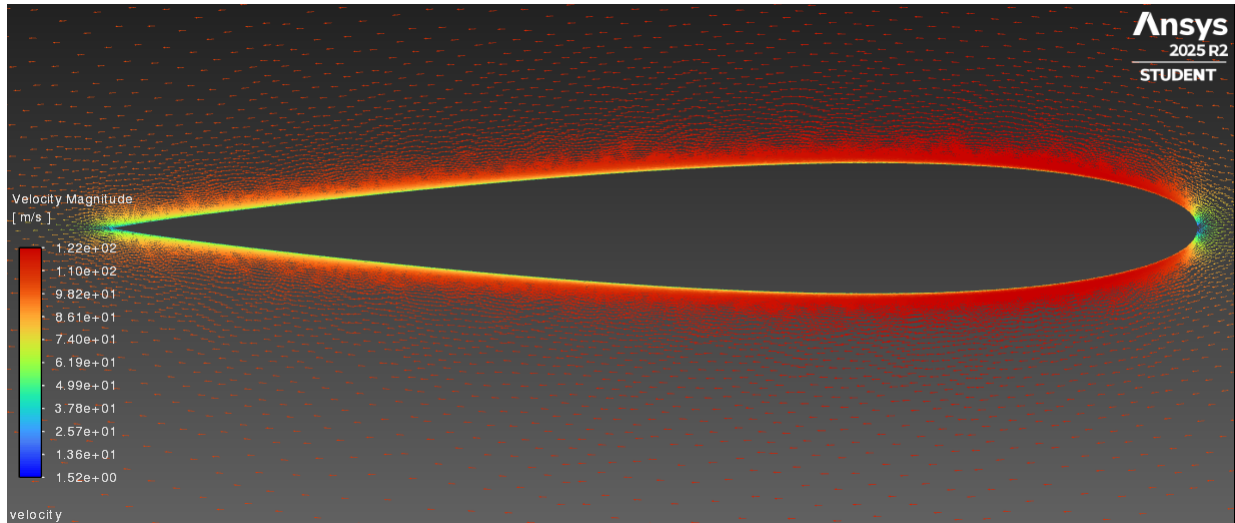


Figure 1.1.2: Velocity Field ($\alpha = 0^\circ$ and $v = 102.9$ m/s)

Looks pretty cool. I was messing around with the vector sizing, and thought this was a good compromise given the limited vision.

Also, due to pure laziness, I won't be providing the lift & drag report convergence plots here (or anywhere else), but I assure you that they did in fact level out to respective steady values, which are tabulated at the end.

Well, the simulated drag (over the airfoil outline) came out to 57.66 N, which seems a bit high. On a more concerning note, the lift came out to 10.86 N, which is absurd! The airfoil is roughly symmetric, and so the lift should be close to zero! Just visually, the fluid speed distribution appears identical above and below the airfoil. I have no idea what went wrong.

The mesh was once again plugged into the CFD solver (coupled, $k - \omega$, 300 iterations) with the appropriate boundary conditions for $v = 171.5$ m/s, and the following results (below) were acquired:

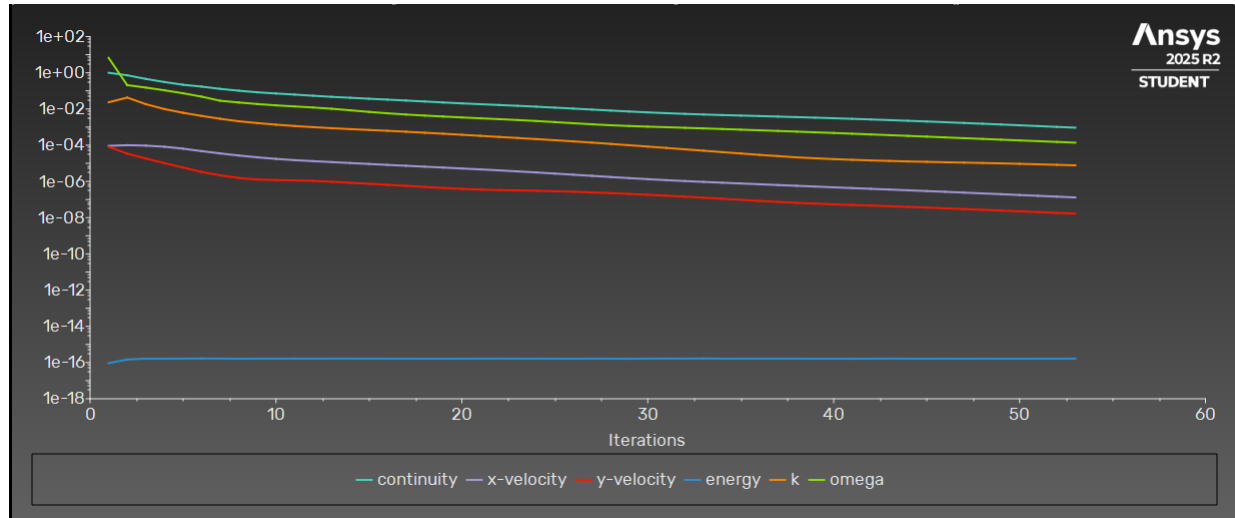


Figure 1.2.1: Residuals ($\alpha = 0^\circ$ and $v = 171.5$ m/s)

Good looking residuals...

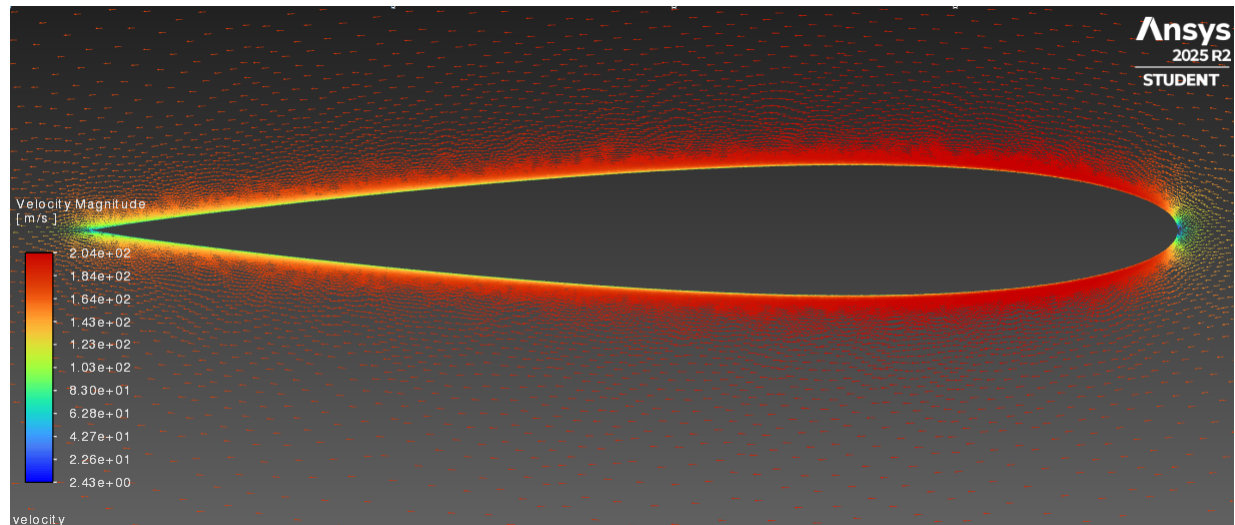


Figure 1.2.2: Velocity Field ($\alpha = 0^\circ$ and $v = 171.5$ m/s)

Nothing out of the ordinary here either...

Of course the same issue showed up here. The drag came out to 149.78 N, and the lift came out to 26.94 N. So something is definitely off, but I just can't place my finger on it. Oh well. Moving on.

Case 2: $\alpha = 10^\circ$

Once again, we need a mesh. Provided a baseline cell size of 0.2 m, in conjunction with inflation (around the airfoil) with 2 layers and edge sizing (0.001 m) to capture the important detail, the following mesh was achieved:

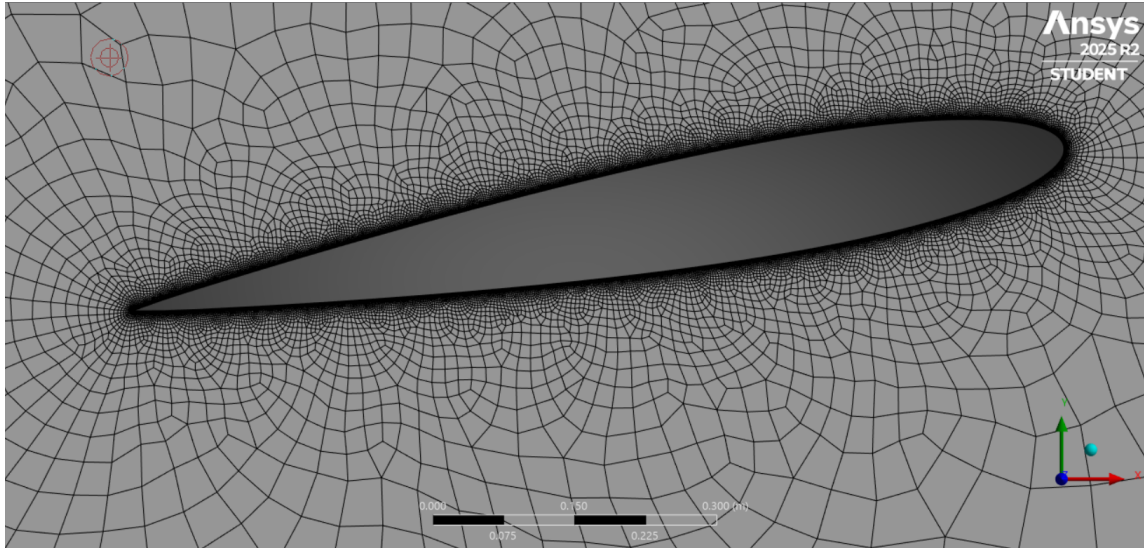


Figure 2.0.0: Mesh Work ($\alpha = 10^\circ$)

The mesh was plugged into the CFD solver (coupled, k - ω , 300 iterations) with the appropriate boundary conditions for $v = 102.9$ m/s, and the following results (below) were acquired.

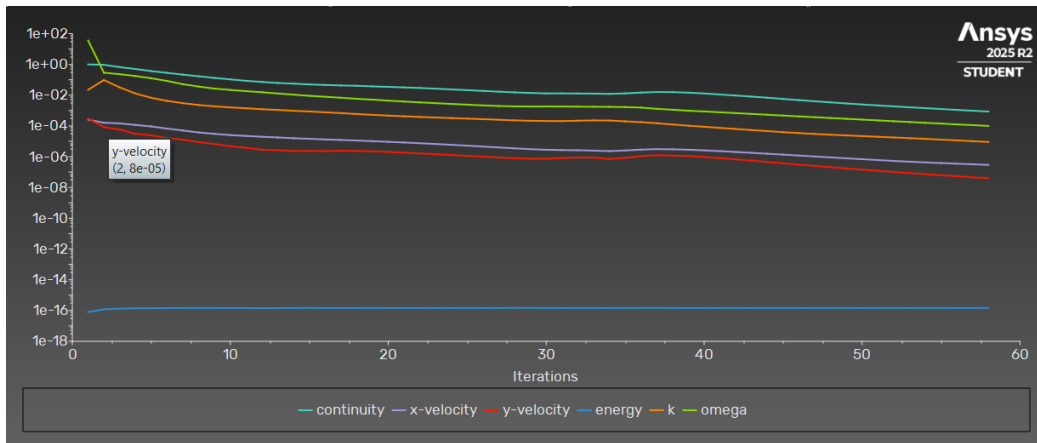


Figure 2.1.1: Residuals ($\alpha = 10^\circ$ and $v = 102.9$ m/s)

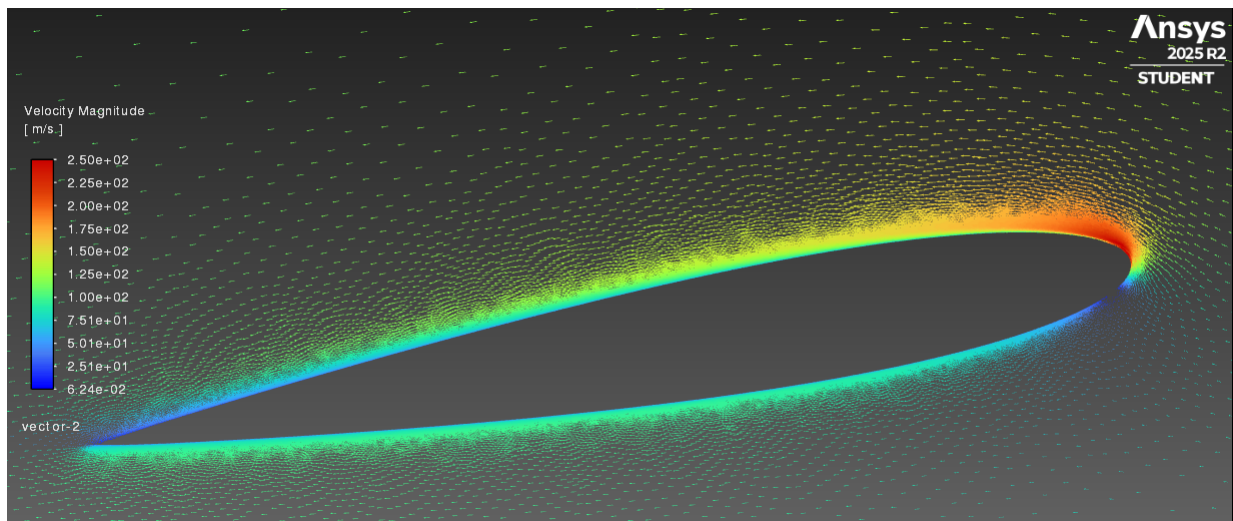


Figure 2.1.2: Velocity Field ($\alpha = 10^\circ$ and $v = 102.9$ m/s)

The mesh was once again plugged into the CFD solver (coupled, $k - \omega$, 300 iterations) with the appropriate boundary conditions for $v = 171.5$ m/s, and the following results (below) were acquired:

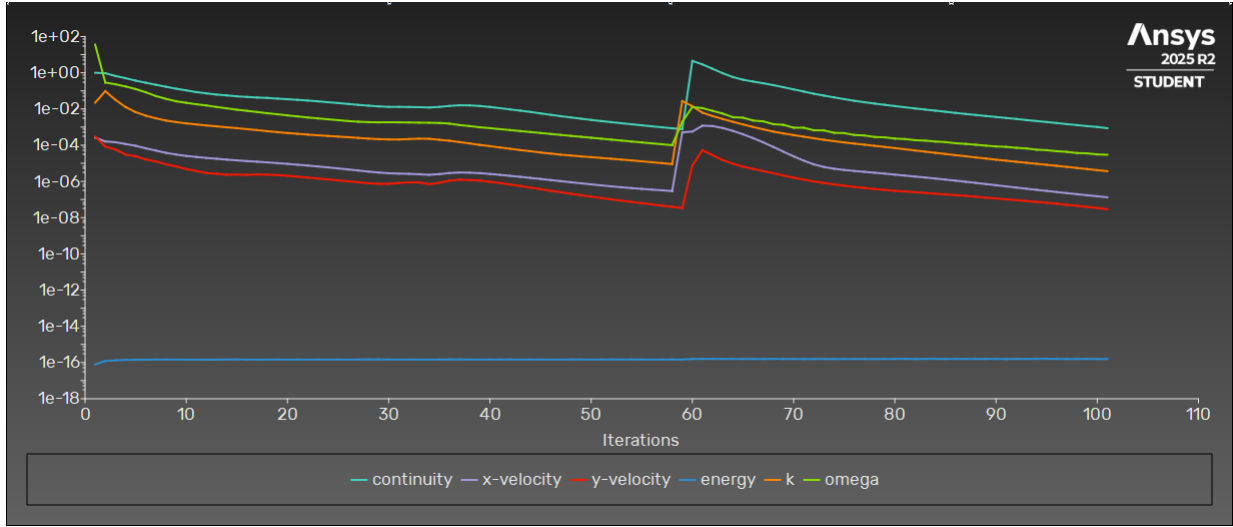


Figure 2.2.1: Residuals ($\alpha = 10^\circ$ and $v = 171.5$ m/s)

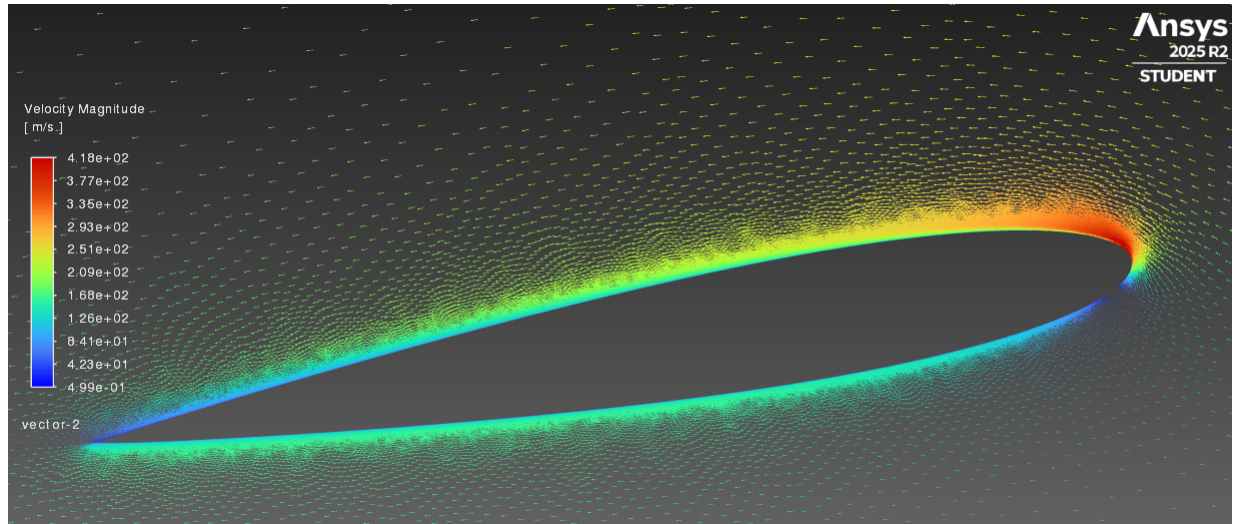


Figure 2.2.1: Velocity Field ($\alpha = 10^\circ$ and $v = 171.5$ m/s)

I've got to say it again. These pictures are always cool to look at. Never gets old... until it does, which will probably be soon.

And don't forget that the force reports are tabulated at the end (spoiler: they aren't as garbage this time).

Case 3: $\alpha = 20^\circ$

For the last time, we need a mesh. Provided a baseline cell size of 0.2 m, in conjunction with inflation (around the airfoil) with 2 layers and edge sizing (0.001 m) to capture the important detail, the following mesh was achieved:

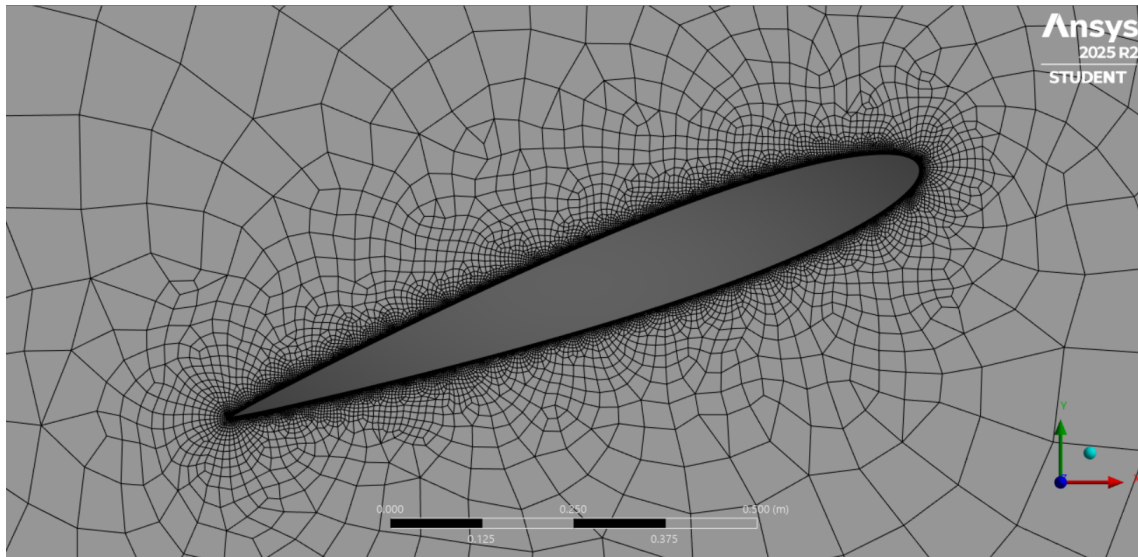


Figure 3.0.0: Mesh Work ($\alpha = 10^\circ$)

The mesh was plugged into the CFD solver (standard, $k - \omega$, 500 iterations) with the appropriate boundary conditions for $v = 102.9$ m/s, and the following results (below) were acquired.

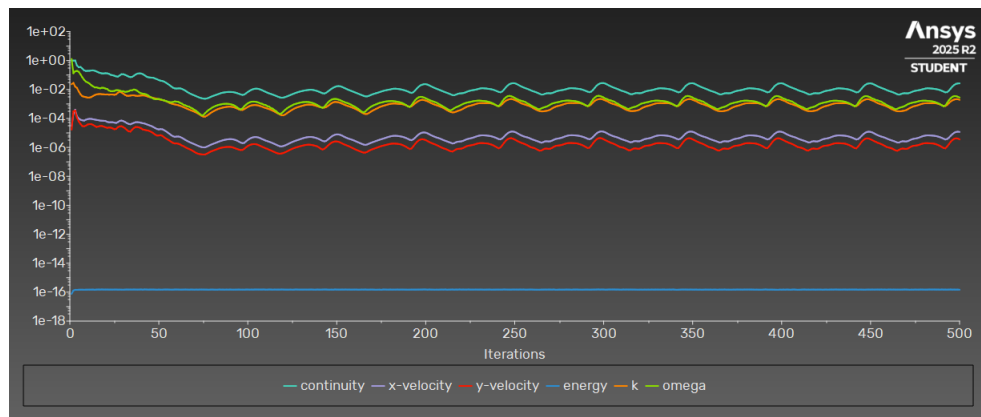


Figure 3.1.1: Residuals ($\alpha = 20^\circ$ and $v = 102.9$ m/s)

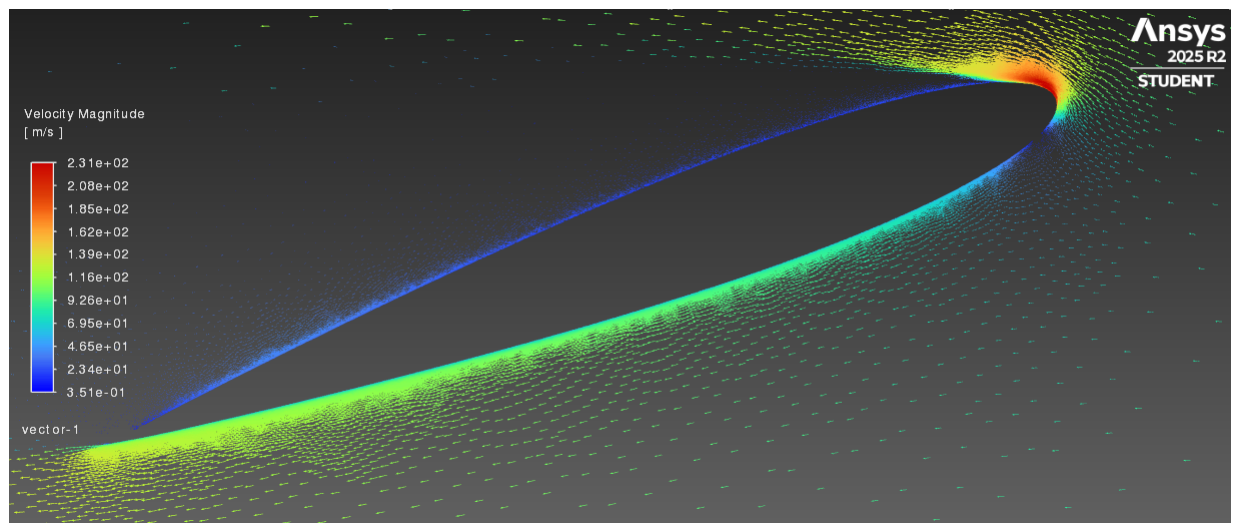


Figure 3.1.2: Velocity Field ($\alpha = 20^\circ$ and $v = 102.9$ m/s)

For the last time, the mesh was plugged into the CFD solver (standard, k - ω , 500 iterations) with the appropriate boundary conditions for $v = 171.5$ m/s, and the following results (below) were acquired.

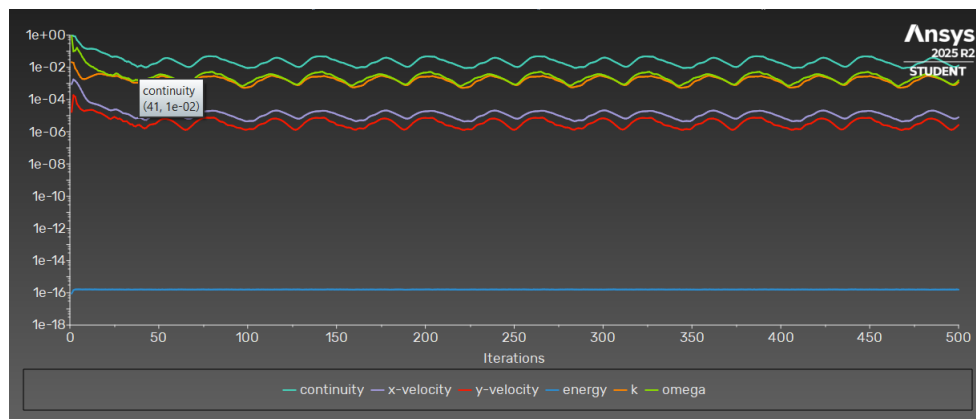


Figure 3.2.1: Residuals ($\alpha = 20^\circ$ and $v = 171.5$ m/s)

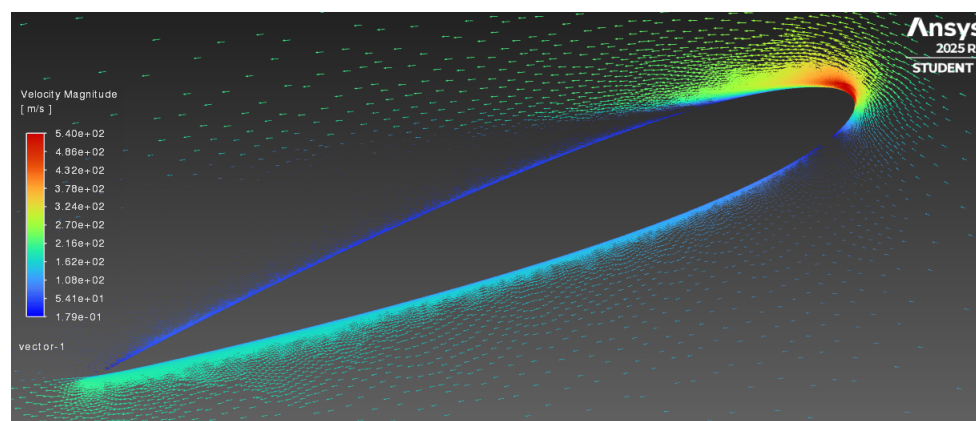


Figure 3.2.2: Velocity Field ($\alpha = 20^\circ$ and $v = 171.5$ m/s)

Tabulated Results

Table 3: Mach 0.3 Sim. Tabulation

α	F_L	F_D
0°	11 N	58 N
10°	6937 N	103 N
20°	5267 N	1562 N

Table 4: Mach 0.5 Sim. Tabulation

α	F_L	F_D
0°	27 N	150 N
10°	19419 N	271 N
20°	22535 N	2636 N

And in case you forgot, or don't want to look back and forth...

Table 1: Mach 0.3 Calc. Tabulation

α	F_L	F_D
0°	0 N	0 N
10°	7150 N	390 N
20°	14200 N	1500 N

Table 2: Mach 0.5 Calc. Tabulation

α	F_L	F_D
0°	0 N	0 N
10°	19800 N	1080 N
20°	39400 N	4140 N

Disregarding the $\alpha = 0^\circ$ case, these numbers line of quite well. I'd call that a moderate success.